Problem Chosen	2025 MCM/ICM Summary Sheet	Team Control Number
B		2508991

From Juno to Global:
Journey to Sustainable Tourism

Due to the increasing degree of globalization, the over-tourism phenomenon is evident in some cities, such as Juneau, Alaska. For such an over-touristy city, it's important to limit the number of visitors in order to achieve sustainable development. The purpose of this paper is to examine the economic, environmental, and social impacts of restricting the number of tourists and regulating taxes and fees, and to provide development options through a combination of non-over-tourism areas.

In order to solve the problem, we mainly build two models: Model 1: sustainable development model based on Simulated Annealing Algorithm(SA) and Analytic Hierarchy Process(AHP). Model 2: Combining Model 1 and Fuzzy Coefficient LP(FCLP) generalization model.

For Model 1: In order to achieve the goal of sustainable development in Juneau, we have established a measurement system, and we evaluate sustainable development through three influencing factors: environmental, social, and economic. The objective function is the composite tourism development index. The constraints are mainly carbon footprint, tax rate, and passenger number constraints, as well as constraints on infrastructure pressures.

Combined with the data provided by Juneau over the years, we consider constraints from the three influencing factors, including but not limited to, the linear relationship between tax rate changes and tourist arrivals, and the non-linear relationship between carbon footprint and tourist arrivals. At the same time, we give the coefficients of the three aspects of the composite index through the AHP method. In the end, we obtained the formula for calculating the comprehensive development index, conducted a sustainable development analysis for the city of Juneau, and found the optimal number of people and tax rate, as shown in Section 3.1.2.

For Model 2: This model combines the method of Model 1 and combines it with the FCLP method. It uses the example of Venice, another over-tourism city, where we divide travelers into three categories. Considering the different geographical locations of Venice, we have certain changes in the economic, social, and environmental considerations, and finally constrained the number of people in five aspects: accommodation, parking spaces, the number of boats, the amount of garbage, and the pressure of scenic spots. Then, by calculating the maximum amount that the number of people can bear, the optimal allocation of tourists in different situations is calculated, which can be seen in Section

In addition, we assume three locations with different tourist volumes and analyze them as a composite index. After a single analysis, we balance over-tourism and non-over-tourism locations and give an allocated plan.

Finally, we performed a sensitivity analysis on the model to verify the stability of the model. After analyzing the pros and cons, we conclude and give recommendations.

Keyword: Sustainable Tourism; Analytic Hierarchy Process; Simulated Annealing Algorithm; The Interval and Fuzzy Coefficient LP Model(FCLP)

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1 Introduction

1.1Background

Global tourism has boomed due to increased globalization and leisure travel. Destinations with unique natural or cultural attractions have leveraged this growth, generating significant economic benefits through spending, job creation, and service industry development [1]. Tourism has also positively impacted residents by fostering community engagement and environmental awareness [2].



Figure 1: Juneau area topographic map and tourist attractions

However, the influx of tourists has also placed increasing strain on local government management. Camatti has detailed the salient symptoms of over-tourism[3], including gentrification, congestion in popular tourist areas, and environmental damage caused by excessive visitor numbers. In Juneau, the melting of icebergs is the most serious problem.

Moreover, the sustainability of tourism in destinations like Juneau hinges on a delicate balance between maximizing economic benefits and mitigating negative impacts. Given the uncertainties of future trends and the dynamic nature of the tourism industry, stakeholders must adopt models that can inform the sustainable management of tourism, ensuring that both the economy and the local community thrive without overexploiting resources[4].

1.2 Restatement of Problems

Our research aims to develop a versatile model for sustainable tourism in a specific region. The model will balance economic benefits with negative externalities, focusing on factors like tourist volume, economic growth, and tourism policies.

Task 1: Model Development. The model will incorporate multiple factors, generating outputs that quantify trade-offs and provide optimization guidelines under various scenarios.

Task 2: Model Extension. The model will be extended to demonstrate its flexibility and applicability to other destinations. Sensitivity analyses will identify key influencing factors.

Task 3: Policy Memorandum. A memorandum will be prepared for the Tourism Commission summarizing projections, evaluating policy interventions, and recommending optimal strategies.

1.3 Our Work

To avoid complicated descriptions and intuitively reflect our work process, the flow chart is shown in the following Figure l :

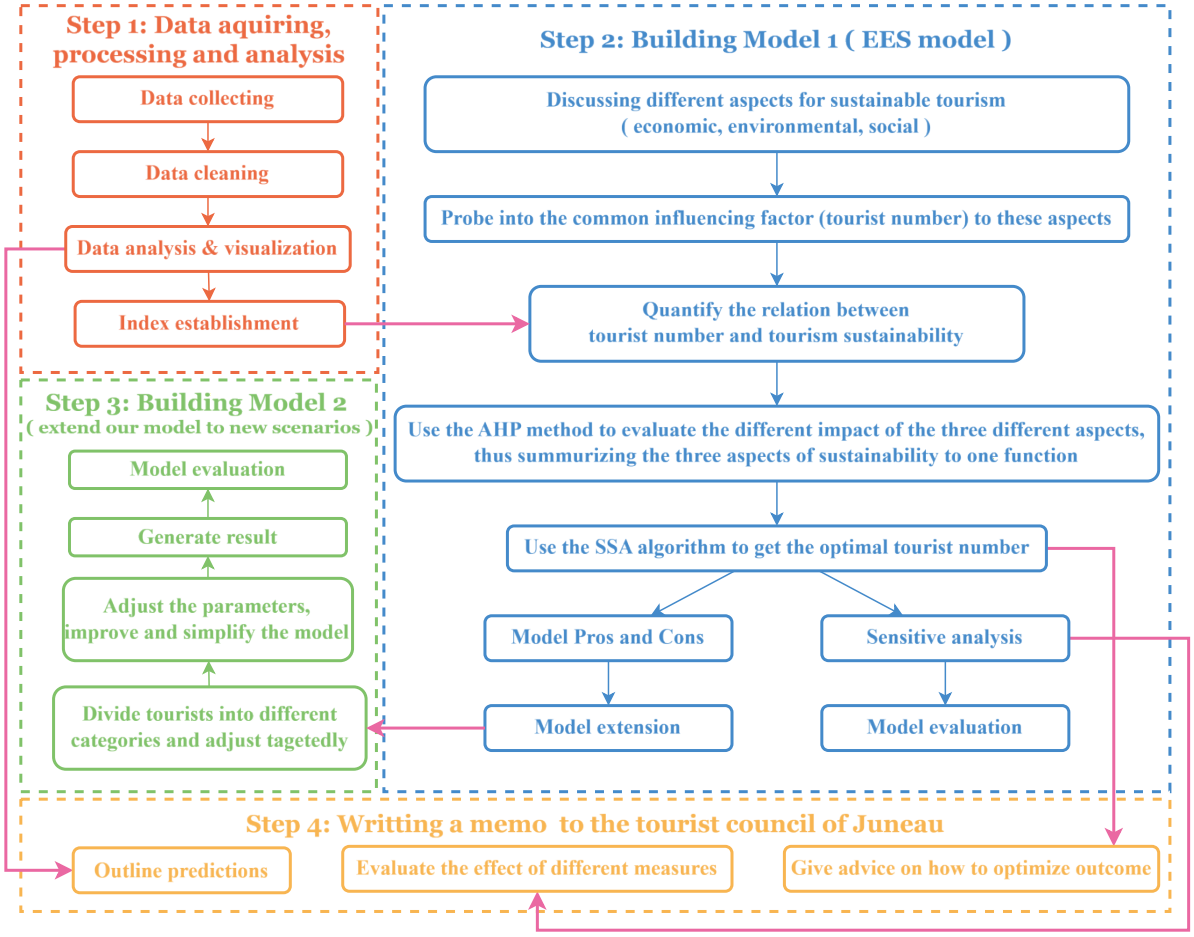


Figure 2: Our framework

1.4 Literature Review

Tourism development involves a delicate balance between the interests of tourists, residents, and local governments. While tourism generates income, it can lead to conflicts over space and resources. Antoci et al. used a Lotka-Volterra model to analyze over-tourism[5], suggesting that controlling local factors and improving resident services may be more effective than tourist restrictions. However, parameter estimation for this complex model presents challenges. He et al., utilizing a feedback-loop model[6], highlighted the importance of wastewater treatment for balancing tourism growth with environmental protection. Recognizing the diversity of tourism destinations, Ceron et al. developed a comprehensive framework based on 15 WTO indicators, providing a standardized tool for assessing and

managing sustainable tourism across different regions.

2 Notations and General Assumptions

2.2 Assumptions

1、Overtourism Assumption: Our model assumes an over-tourism scenario where visitor numbers have reached a level that maximizes economic benefits.

2、 Tax Rate and Visitor Number Relationship: Existing research has shown that a 3% increase in tax rates leads to a 5.4% decrease in visitor numbers. We assume a similar relationship holds for the Juneau region.

3、 Seasonality Assumption: Our model considers annual visitor numbers (x), ignoring seasonal fluctuations in monthly visitor counts.

4、Social Impact Assumption: The model assumes a positive correlation between visitor numbers and the decline in local residents' quality of life. For quantification, we assume this indicator is directly proportional to x.

2.1 Notations

Figure 3： Important parameters used in the paper

Symbol	Description
M	Comprehensive Development Index
I_{eco}	Economic Impact Factor
I_{env}	Environmental Impact Factor
I_{sco}	Social Impact Factor
x	Number of tourists per year
t	Consumption tax rate
t^*	Tax rate increase under the base year
k, b	Parameters of the fitting curve of tourism income and number of tourists
$\varphi(x)$	Carbon footprint per year
A_j	Residents' perception of tourism impact

3 EES Full-factor Tourism Development Model

3.1 The Inspiration for The EES Model

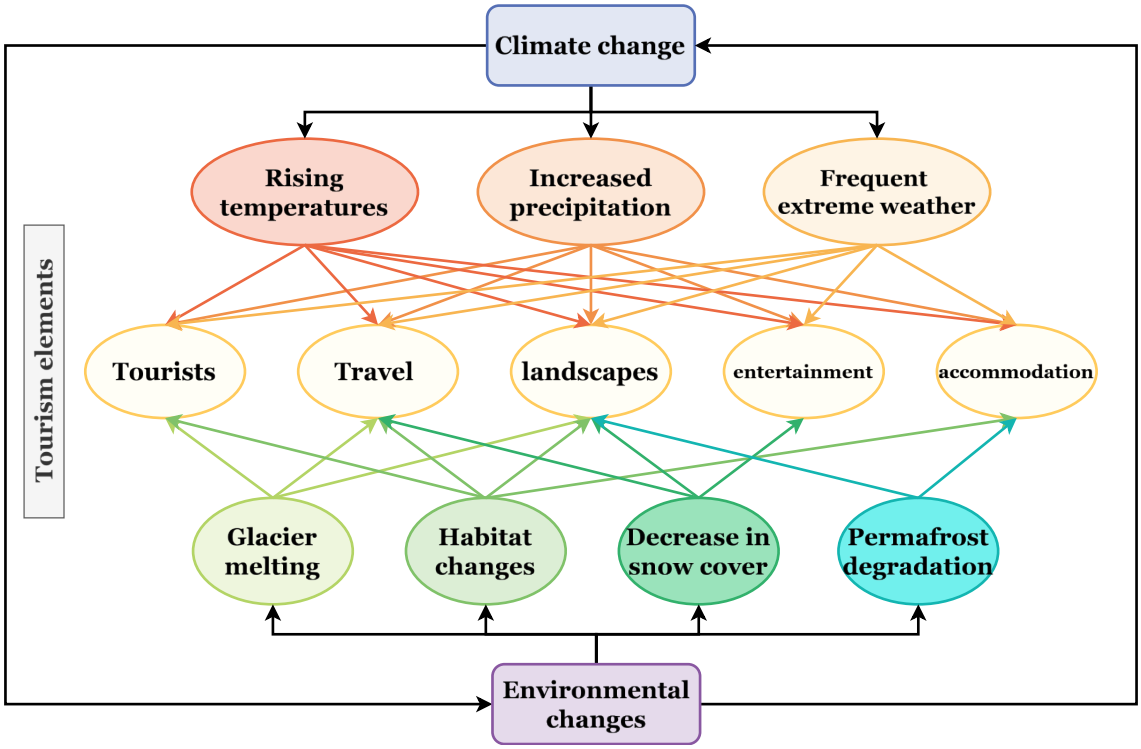


Figure 4: How climate and environmental change affect tourism elements

This study proposes a sustainable tourism development model that integrates economic, ecological, and social dimensions. Drawing on complex systems science, ecological economics, and sociology, it highlights the interconnectedness of tourism destinations and aims to guide balanced development while ensuring equitable social outcomes.

3.2 Assessment of Economic, Ecological, and Social Factors

3.2.1 Macro Overview

So overall, our mathematical model is grounded in the Analytic Hierarchy Process (AHP). By adjusting the annual number of tourists and local tax rates based on known data, we have quantitatively analyzed three key impact factors:

- **Economic impact factors I_{eco} :** Tourism revenue changes can be assessed by comparing the ratio of total revenue in the target period to that of the base period, considering tax-tourist equilibrium and shifting revenue benchmarks.

$$I_{eco} = \frac{(1+t^*)\,(kx(t^*)+b)}{(kx(0)+b)}$$

- **Environmental impact factors I_{env} :** A linear equation was developed to model the relationship between tourist numbers and environmental impacts, such as waste generation

and transportation emissions. This model was adjusted for global warming effects and normalized for comparability.

$$I_{env} = 1 - \frac{\varphi(x) - \varphi_{\max}}{\varphi_{\max}} - g$$

$$\varphi(x) = x\left(\sum e_j\right) \times n \ \ (j = 1, 2 .....)$$

◆ **Social impact factors I_{soc}:** The model incorporates resident satisfaction as a key social factor by establishing a positive correlation between resident satisfaction and tourist numbers.

$$I_{soc} = 1 - \lambda \frac{\sum_{j=1}^{n} \frac{A_j}{n} - k^2}{1 - k^2} x$$

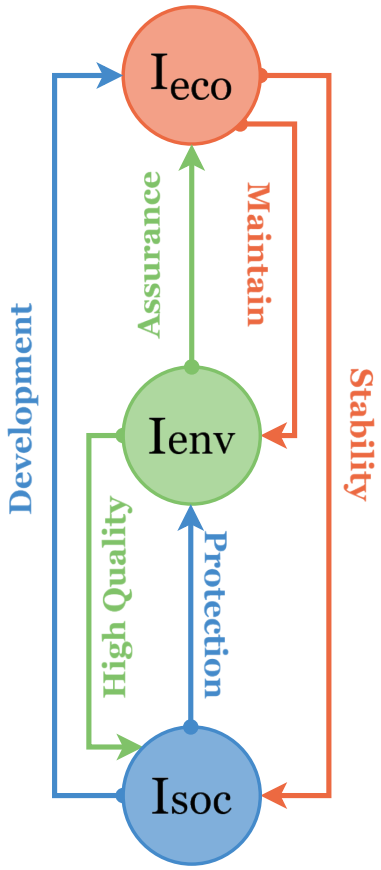


Figure 5: Connections between the three objects

A comprehensive tourism development index was derived using the AHP and weighted summation, enabling data-driven policy recommendations. To analyze the industry's evolution, the study focuses on the relationship between tourist numbers and the government's consumption tax rate. This will be elaborated in detail below

3.2.2 The Economic Impact Factor

This indicator has no dimension and reflects the level of economic benefits generated by the tourism industry in a specific region. The higher its value, the better the economic development trend generated by the tourism industry in the local area and the more benefits it has. As shown in the formula (), this expression can be interpreted and used in two ways, each with its own advantages in different situations.

$$I_{eco} = \frac{(1+t)\,(kx(t)+b)}{(1+T_0)\,(kx(T_0)+b)}$$

1、Basic ideas and logic

Firstly, given the abundance of historical data, we can interpret the code in the equation as follows:

T_0 represents the tax rate in the base year, which can be set to 0.09 for Juneau. The term $kx+b$ represents the total expenditure of tourists, which is equivalent to the total revenue of the region. This is approximated by revenue from local tourism industries, based on accurate data. We also obtained the total number of tourists arriving in Juneau via ferry, cruise ship, and air (Figure 6).

Figure 6: Tourists entering Juneau by different entry routes and tourism revenue

Year	Cruise Ship Visitation	Ferry Passenger	Air Passenger	Hotel & Motel Gross Business Sales	total earnings
2014	953100	72187	307742	32071	78387078
2015	982500	65101	331079	33439	84968566
2016	1015100	59194	339279	34677	84506190
2017	1072300	57144	345454	35603	88790372
2018	1151100	53920	358388	35906	86018238
2019	1305700	41559	365349	37496	103225389
2020	37	10987	154292	19077	62723855
2021	115800	25299	284039	37829	78383883
2022	1167194	35683	359312	53740	119520965
2023	1638902	41469	354905	59158	134631332

Furthermore, upon thorough analysis and comparison, we have found a highly significant positive correlation between the number of tourists and tourism-related revenue, with a Pearson correlation coefficient of 0.993901. A linear regression analysis was conducted, and the resulting regression line, along with the data visualization, are presented in [Figure X].

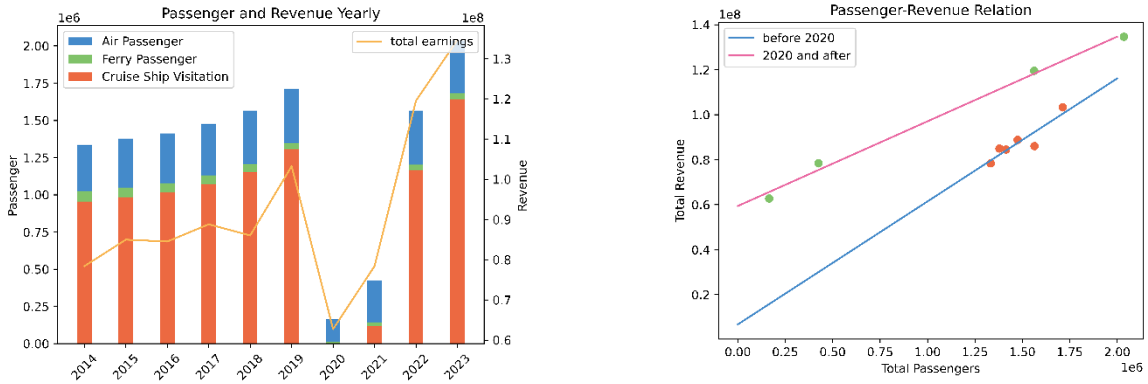


Figure 7: Visitor data visualization and the effect of straight-line fitting

Upon analysis, it is evident that the two fitted regression lines correspond to the data before and after 2020. As we know, the COVID-19 pandemic hit the globe in early 2020, significantly impacting regions like Juneau that heavily rely on tourism. The pandemic led to increased maintenance costs for cruise ships, bankruptcies of tourism companies, decreased government budgets for tourism, higher sunk costs, disrupted capital flow, and a decline in economic vitality, all of which contributed to a substantial drop in tourism revenue in Juneau. However, with the gradual return of tourists, revenue levels have rebounded. Moreover, the tourism market has demonstrated increased resilience and adaptability post-pandemic. Consequently, for our predictive and evaluative models, we have adopted the linear equation derived from the post-2020 data.

$$f(x) = kx + b, \quad k = 37.6488 \; b = 59397421.1857$$

Then in the numerator, it is the variable tax rate t that we set and the number of passengers x affected by it (see Constraints for details.). Similarly, we multiply the local trademark price income by a tax rate surcharge to obtain the total income. By comparing the numerator and denominator, we can get the corresponding parameters.

2、Extension of ideas

Alternatively, we propose a second interpretation that can be employed when data is limited or under specific circumstances. In this scenario, we choose to disregard the known consumption tax from previous years by setting T0 to 0. Consequently, the denominator can be interpreted as the total transaction value contributed by consumers in the base year. We argue that imposing additional taxes on this basis does not compromise the overall validity of the model, as the fundamental relationship between commodity prices, taxes, and total revenue remains unchanged.

However, to accommodate this new assumption, we need to adjust the relationship of x(t) and denote it as x1(t). Nevertheless, the fitted relationship of kx+b remains unchanged. Essentially, this approach allows us to consider an additional tax rate, denoted as t*, and subsequently calculate the corresponding value of I.

Furthermore, we can establish a connection between these two expressions using the following equation.

$$\frac{(1+t)\left(kx(t)+b\right)}{(1+T_0)\left(kx(T_0)+b\right)}=\frac{(1+t^*)\left(kx_1(t^*)+b\right)}{kx_1(0)+b}$$

Because of the special nature of the relationship between x and t (equation 1), we know:

$$\frac{x(T_0)}{x_1(0)}=\frac{x(t)}{x_1(t^*)}$$

So according to the relationship between the coefficients of b, we can solve:

$$t^*=\frac{t-T_0}{1+T_0}\ \textbf{aka.}\ t=(1+T_0)t^*+T_0$$

For the sake of convenience and scientific rationality of modeling, we adopt the second explanation to deal with the model in the Juneau region.

3.2.3 Environmental Impact Factors

1、Formula Introduction

In today's rapidly developing world, economic growth at the expense of environmental degradation is unsustainable. Therefore, it is crucial to assess the environmental conditions of tourist areas and the level of disturbance caused by tourists. So we define a special indicator.

$$I_{env} = 1 - \frac{\varphi(x) - \varphi_{\max}}{\varphi_{\max}} - g$$

$$\varphi(x) = x\left(\sum e_j\right) \times n \ \ (j = 1, 2.....)$$

Where $\varphi(x)$ represents the total carbon emissions generated by all tourists in a year, e represents the carbon footprint generated by a tourist per day in the Juneau region, n represents the average length of stay of each tourist in the Juneau region, and g is a constant parameter related to glacier melting. Meanwhile, $\varphi(x)$ represents the upper limit of carbon footprint emissions that the Juneau region can withstand annually, which is also the national standard carbon emission baseline. By using the Min-Max normalization method, the dimensional carbon footprint is mapped to the interval [0,1], which also establishes a negative correlation between the target indicator and the number of tourists x.

2、About Carbon Footprint

The measurement of carbon footprint is a crucial indicator for the internationally advocated goals of 'carbon neutrality' and 'carbon peak'. Numerous international scholars have made significant contributions to the quantification and measurement of carbon footprints. María-Ángeles and his colleagues[8], utilizing the input-output model, have separately measured direct and indirect carbon emissions. The formula is as follows.

$$CF_r = \hat{e}\,(I - A^{*d})^{-1}\,\widehat{C_h} + \hat{e}\,(I - A^{*t})^{-1}\left(A^m (I - A^{*d})^{-1}\,\widehat{C_r} + \widehat{C_r^m}\right)$$

$$CF_v = \hat{e}\,(I - A^{*d})^{-1}\,\widehat{C_v} = \widehat{e^*}\,(I - A^{*t})^{-1}\left(A^m (I - A^{*d})^{-1}\,\widehat{C_v} + \widehat{C_v^m}\right)$$

In the equation, e is the direct carbon emission coefficient or vector. A is the input-output matrix, which describes the input-output relationship between different sectors. I is the labor input matrix. d is the final demand vector. $\widehat{C}$ is the consumption matrix.

We try to imitate and learn from these predecessors, using relevant income-output model data to simulate and try to construct the carbon footprint constant of our own model. In addition, since most tourists come to Juneau by cruise, it is estimated that their average stay is about 5 days. We also consulted the information to supplement the international standard carbon emission bottom line. The comprehensive data obtained:

$$\varphi(x)\big/\varphi_{\max} \approx x\left(\sum e_j\right) \times \frac{n}{\varphi_{\max}} = \frac{5 \times 0.44}{18 \times 365 \times 10^4} \approx 3.3 \times 10^{-8}$$

3、About glacier melting

A significant amount of media has suggested that a surge in tourist numbers is a definitive factor in accelerating glacial melt. However, our experimental and investigative findings indicate that the accelerated melting rate of the Juneau Icefield over the past two decades is primarily attributable to the broader trend of global warming rather than the volume of tourists,

as we've been discussing. So we assign a constant g to the fitting formula, and after data analysis, we get $g=1.2$.Studies show a link between glacial accumulation and summer temperature, as well as snowfall. We compared recent snowfall and rainfall data in Juneau with tourist numbers. The data is as follows.

Figure 8: Data related to the fitted line

	Slope of a line	Line intercept	Correlation coefficient
Highest temperature of the year	0.000001	81.776849	0.04462
Year-round median temperature	0.000001	-1.414726	0.087519
Lowest temperature of the year	0.000002	-1.414726	0.051862
Snowfall all year round	-0.000004	77.576392	0.048617

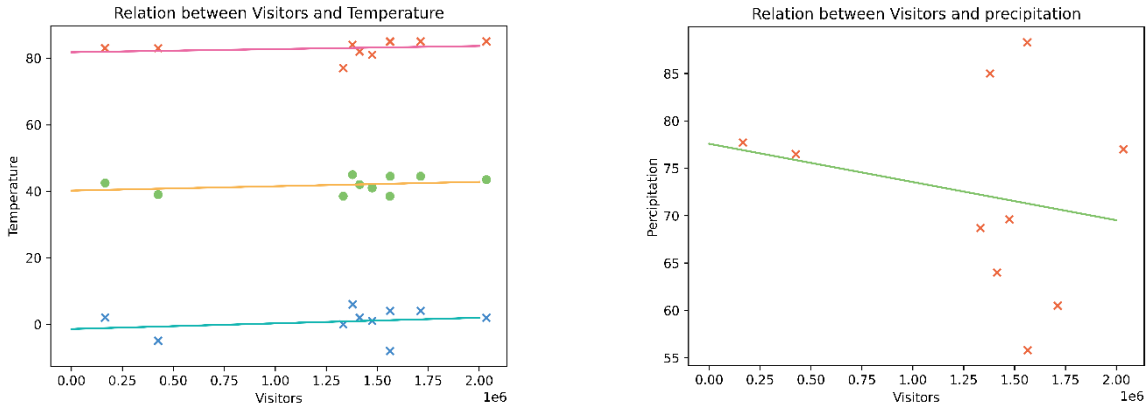


Figure 9: Fitting of annual rainfall and snowfall to the tourists' number

3.2.4 Social Impact Factors

1、About The Formula

The socio-cultural impacts of tourism, a significant yet challenging aspect of tourism's overall influence, encompass all changes in a destination's society and culture that can be attributed to tourist activities or tourism industry operations. We define a social impact factor as follows。

$$I_{soc}=1-\lambda\frac{\sum_{j=1}^{n}\frac{A_j}{n}-k^2}{1-k^2}x$$

x denotes the number of passengers, n signifies the total number of questions subject to evaluation, λ is a parameter that balances the first two indicators and A_j signifies the degree to which individuals perceive the j-th question, as defined by:

$$A_j = D_j \times E_j = \left(\sum_{i=1}^{k} a_{ji} d_{ji}\right) \times \left(\sum_{i=1}^{k} a_{ji} e_{ji}\right)$$

Where D_j notes the magnitude of alteration in individuals' perception of the tourism-induced impact[9], E_j signifies the extent to which individuals accept these occurrences, k represents the number of scale categories, a_{ji} denotes the proportion of respondents selecting the i-th category of the scale when confronted with the j-th item, d_{ji} denotes the perceived change score associated with the i-th category of the scale for the j-th item, and e_{ji} denotes the acceptance score associated with the i-th category of the scale for the j-th item. In this research, $n = 1$,$\lambda = 1.0 \times 10^{-7}$ and $k = 5$.

After the chronological survey, the data of matrices $\boldsymbol{a}$, $\boldsymbol{d}$ and $\boldsymbol{e}$ were obtained. The stronger the perception of the problem, the greater the corresponding scale value.

$$D = [.43\ .30\ .12\ .10\ .05] \times [5\ 4\ 3\ 2\ 1]^T = 3.96$$

$$E = [.32\ .17\ .29\ .11\ .11] \times [5\ 4\ 3\ 2\ 1]^T = 3.73$$

So the coefficient of x is approximately $1.0 \times 10^{-7} \times \dfrac{3.73 \times 3.96 - 25}{1 - 25} \approx 4.3 \times 10^{-8}$



Figure 10: Juneau residents' attitudes toward tourism development over the years

2、Explanation

As the number of tourists increases, inevitable conflicts arise with local residents over spatial, material, and cultural resources. The daily lives of local residents may also be disrupted, potentially leading to changes in ethical and moral values and social instability. Therefore, we utilize a negative correlation with tourist numbers as a metric, which is then positively transformed for measurement[10].

3.3 Analysis of Constraints

Based on the sophisticated and coherent model outlined above, we have incorporated four constraints into the three fundamental definitions. These constraints have been meticulously

developed through a comprehensive analysis of relevant policies, empirical data, and existing literature.

$$\begin{cases} 0 < x < N_{\max} & (1) \\ d(\log_{(1-5.4\%)} x) = d\left(\dfrac{t}{3\%}\right) & (2) \\ I_{env} < K & (3) \end{cases}$$

In Equation (X), we define the tourism resource carrying capacity[11] as the maximum number of tourists a destination can support based on its resources and infrastructure. This capacity is categorized into five components: tourism space, water, land, vegetation resources, and tourist facilities.Excluding tourism resources, development indicators vary under different carrying capacities. We examine consumption power to understand the tourism resources' carrying capacity in this model.

$$N_{\max} = \sum_{i=1}^{n} \left(\frac{T}{t} \times M \times N \right)$$

where i represents the category of the vehicle, especially ferries or cruises, T represents the time required for vehicles to go back and forth, t represents the work time for traffic tools average daily, N represents the number of tourist vehicles that can carry, and M represents the number of available means of transport.

Secondly, research has shown that a 3% increase in tourism tax can lead to a 5.4% decrease in tourist demand[12]. Based on this finding, the differential equation can be listed to describe the relationship between the number of tourists x and the tax rate t. After solving, we can get the following results.

$$x = C \cdot \left(0.946^{\frac{t-1}{0.1}}\right)$$

Finally, as equation mentioned above, ecological carrying capacity can be further divided into several aspects: water quality indicators, air quality indicators, soil quality indicators, and energy consumption indicators, etc.

$$UWECC = W_P / P_W \quad UAECC = E/S \quad USECC = S/D$$

$$T = \sum_{i=1}^{n} (a_i \times n) \times e_i \quad S_E = \sum_{i=1}^{n} S_i \times N_i \times R_i / B \quad E = \sum_{j=1}^{n} A_j \times r_j \times y_j$$

Among those parameters, what needs to be said is that A_j represents all kinds of biological production areas, r_j represents equilibrium factors, and y_j represents production actors. E represents the atmospheric environment quality of scenic spots, S represents the national standards for atmospheric environmental level and D represents national standards.

After meticulously collecting data on these indicators in the Juneau region, we selected the minimum value of these four indicators as K, representing the upper limit of the region's

ecological carrying capacity. Therefore, the objective function I_{env} in our model, which represents the environmental impact factor, is constrained by K. In the parameter optimization process, considering the principle of sustainable tourism, I_{env} should not exceed K.

3.4 Analytical Hierarchy Process for Evaluating the Results

By utilizing the aforementioned three modeling algorithms, we can construct a comprehensive development index. The coefficients a, b, and c within this index will be scientifically determined through the analytic hierarchy process (AHP).

$$M = \eta_1 I_{eco} + \eta_2 I_{env} + \eta_3 I_{soc}$$

First, we consulted the relevant information on "economic, social and ecological coupling system" and constructed a judgment matrix using the hierarchical analysis method. The heat map is shown below (). First, a consistency test is performed $CI = (\lambda_{\max} - n)/(n-1)$.from the table, we know that RI is 0.52, and its consistency test index $CR < 0.1$ .s

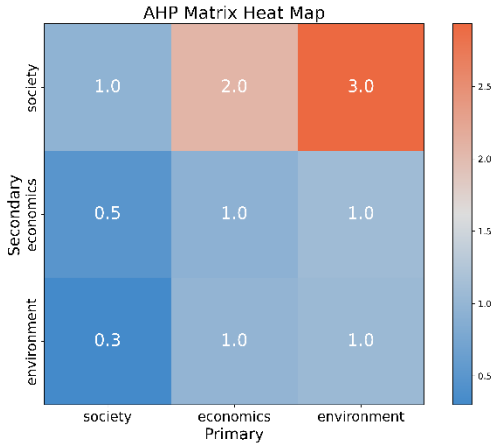


Figure 11: AHP Matrix heat Map

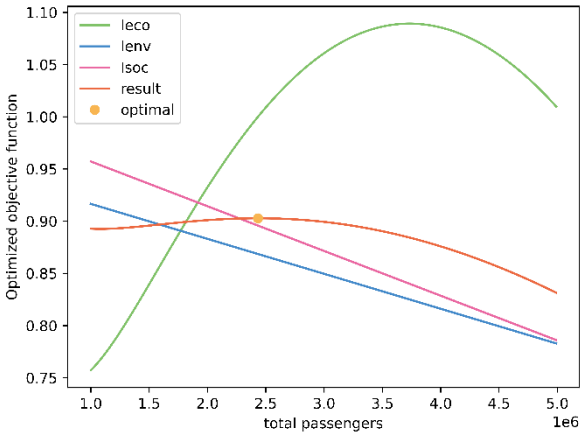


Figure 12: Core variable curve chart

By normalizing each column of the pairwise comparison matrix in the analytic hierarchy process (AHP) and calculating the average, we can derive the corresponding weight coefficients.

$$\eta_1 = 0.21,\ \eta_2 = 0.55,\ \eta_3 = 0.24$$

Therefore, we can conclude that the comprehensive development index M and its three sub-indices show a curve in relation to the number of tourists. After optimization using the Simulated Annealing Algorithm (SSA), the optimal solution, or the number of tourists, is found to be $x_{opt} = 2433521$ people. Compared to the base year reference system, the increment in the tax rate differs from 1 by a magnitude of 10^{-6}, which suggests that the tax rate should remain unchanged at 1.76%. Additionally, we observed that when the number of tourists is less than x_{opt}, M does not vary significantly with x. However, once this threshold is exceeded, M decreases rapidly, leading to a decline in tourism quality and sustainable development levels. Therefore, local governments can maintain a macroscopic balance under appropriate constraints to ensure that the number of tourists does not exceed the aforementioned range, and there is no need to be overly concerned about the exaggerated negative impacts publicized by the media. Nevertheless, governments should maintain a high level of tourism investment to address issues

related to environmental and infrastructure maintenance.

4. The Promotion of The Model

While Model 1 cannot be directly applied to cities like Venice due to unique factors like whale watching and glacier scenery, it can be adapted. By analyzing hypothetical locations with varying popularity within a city, we can achieve a balance between attractions and promote sustainable development.

4.1 Analysis of The TCC in Venice

4.1.1 Objective Function

Venice, another over-touristed city, requires a tailored approach. TCC (Tourism Carrying Capacity) is crucial for sustainable development. Existing threats, identified through monitoring efforts, include large boat traffic and inappropriate construction. We'll adapt our model accordingly. Therefore our model for Venice excludes glacier impacts. Given its developed transport sector, we'll analyze the effects of various transportation modes. We categorize visitors as: hotel guests, luxury hotel guests, and day-trippers.

1、To simplify the economic factor, we constrain visitor numbers in Venice by the available bed capacity across different accommodation types, reflecting the linear relationship between economic growth and visitor numbers observed in Model 1

2、Environmental factors include cruise ship arrivals and waste disposal. Visitor numbers influence these factors. Our model aims to minimize environmental impact by reducing cruise ship arrivals and optimizing waste management.

3、Social factors indicate how crowded Venice's famous attractions are, and in this model, we use Piazza San Marco as an example.

$$F = opt[I_{eco}, I_{env}, I_{soc}]$$

In the case of Venice, the survey found that there was a strong correlation between the tourists and the above three effects, so we optimized the model to the following model:

$$F = \max_{x} \sum_{i=1}^{3} c_i x_i$$

where Ci(i=1,2,3) represents the coefficient corresponding to the number of passengers in different types in terms of environmental impact. Xi(i=1,2,3) indicates different categories of tourists.

4.1.2 Constraint

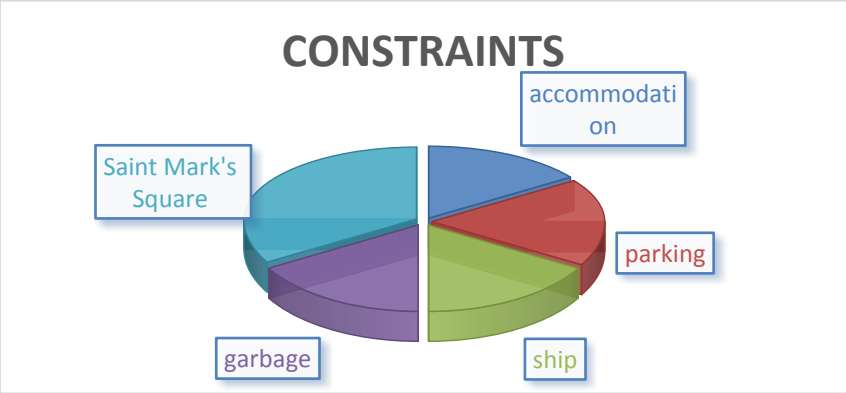


Figure 13: Constraints on tourism in Venice

Taking into account the nature of Venice as a "water city", we took the following six aspects as constraints for this planning model:

1、Accommodation

Venice received 19 million tourists in 2019, with a maximum of 18,000 hotel beds available. Day trippers do not require accommodation, so the accommodation constraint is as follows:

$$\begin{cases} x_1 \leq D_{1max} \\ x_2 \leq D_{2max} \end{cases}$$

where D_{1max} indicates the maximum number of beds that can be provided by a hotel per day, and D_{2max} indicates the maximum number of beds that can be provided by non-hotel accommodations such as hotels per day. Taking into account the fact that there are passengers staying for more than one day, we will consider these two as 70% of the full class time.

2、Parking

Even though Venice is a city on the water, since almost 80% of visitors to Venice are day trips, we have to consider the impact of parking spaces on the number of tourists.

$$\alpha_{11}x_1 + \alpha_{12}x_2 + \alpha_{13}x_3 \leq M_1$$

where a represents the influence coefficient of the number of parking spaces on x_i (i=1,2,3). M_1 indicates the maximum number of parking spaces (mainly considering the number of parking spaces in the historic center of Venice and Mestre.

3、Ship

In Venice, public transport boats are the main means of travel. The number of boats influences both the ease of tourist movement and the flow to key attractions.

$$\alpha_{21}x_1 + \alpha_{22}x_2 + \alpha_{23}x_3 \leq M_2.$$

Here, aaa represents the influence coefficient of the number of ships on x_i (i=1,2,3) M_2 denotes the maximum passenger capacity of a vessel, primarily considering public transport boats on the Grand Canal.

4、Garbage

Increased tourism will significantly raise urban waste, impacting the city's sustainable development. Venice must limit tourist numbers to prevent waste from exceeding its capacity.

$$\alpha_{31}x_1 + \alpha_{32}x_2 + \alpha_{33}x_3 \leq M_3.$$

Where a represents the influence coefficient of garbage quantity on x_i (i=1,2,3). M_3 indicates the amount of waste generated by visitors to Venice on a daily basis.

5、Saint Mark's Square(SMS)

Based on the impact of iceberg melting on Juno's tourist numbers, we can predict the effect of congestion at Venice's attractions, like Piazza San Marco. Thus, we need to limit the daily visitor capacity of Piazza San Marco, as follows.

$$\alpha_{41}x_1 + \alpha_{42}x_2 + \alpha_{43}x_3 \leq M_4.$$

Where a represents the influence coefficient of the maximum number of people who are restricted from going to the attraction on x_i (i=1,2,3).M_4 indicates the maximum daily capacity of Piazza San Marco.

4.1.3 Model Summary

By summarizing the above objective functions and constraints, we can obtain the following optimization model (LP):

$$\begin{cases} \max\limits_{x} \sum\limits_{i=1}^{3} C_i\, x_i \\ x_1 \leq D_{1max} \\ x_2 \leq D_{2max} \\ \alpha_{11}x_1 + \alpha_{12}x_2 + \alpha_{13}x_3 \leq M_1 \\ \alpha_{21}x_1 + \alpha_{22}x_2 + \alpha_{23}x_3 \leq M_2 \\ \alpha_{31}x_1 + \alpha_{32}x_2 + \alpha_{33}x_3 \leq M_3 \\ \alpha_{41}x_1 + \alpha_{42}x_2 + \alpha_{43}x_3 \leq M_4 \\ x_1, x_2, x_3 \geq 0 \end{cases}$$

However, since these data parameters are not stable and constant, it is difficult to find a definite parameter, so we consider the fuzzy interval model. Figure. We use alpha cutting at this point to make further estimates:

$$Section.1(\lambda)\begin{cases} \max\limits_{x} \sum\limits_{x}^{n} \gamma_j\,(\lambda)x_j \\ \sum\limits_{i=1}^{n} \beta_{ji}\,(\lambda)x_j \leq \varepsilon_i(\lambda) \\ x \geq 0 \end{cases} \qquad Section.2(\lambda)\begin{cases} \max\limits_{x} \sum\limits_{x}^{n} \delta_j\,(\lambda)x_j \\ \sum\limits_{i=1}^{n} \alpha_{ji}\,(\lambda)x_j \leq \eta_i(\lambda) \\ x \geq 0 \end{cases}$$

Combining the above methods, we have further improved the model

$$\begin{cases} \max\limits_{x} \overline{c_1} x_1 + \overline{c_2} x_2 + \overline{c_3} x_3 \\ x_1 \le \overline{\mathrm{D_{1max}}} \\ x_2 \le \overline{\mathrm{D_{2max}}} \\ \overline{\alpha_{11}} x_1 + \overline{\alpha_{12}} x_2 + \overline{\alpha_{13}} x_3 \le \overline{\mathrm{M_1}} \\ \overline{\alpha_{21}} x_1 + \overline{\alpha_{22}} x_2 + \overline{\alpha_{23}} x_3 \le \overline{\mathrm{M_2}} \\ \overline{\alpha_{31}} x_1 + \overline{\alpha_{32}} x_2 + \overline{\alpha_{33}} x_3 \le \overline{\mathrm{M_3}} \\ \overline{\alpha_{41}} x_1 + \overline{\alpha_{42}} x_2 + \overline{\alpha_{43}} x_3 \le \overline{\mathrm{M_4}} \\ x_1, x_2, x_3 \ge 0 \end{cases}$$

4.1.4 Result and Analysis

Figure 14: coefficient interval and Optimal number of passengers

The fuzzy parameter interval of the coefficient						
	X1_central	X1_high	X2_central	X2_high	X3_central	X3_high
C	210	230	160	180	60	80
Hotel	1	1	0	0	0	0
Non-Hotel	0	0	1	1	0	0
Parking	0.27	0.355	0.27	0.355	0.75	0.825
Ship	1	1.11	1	1.11	1	1.11
Garbage	0.755	0.823	0.67	0.729	0.2	0.22
SMS	0.5	0.6	0.3	0.4	0.7	0.8

Estimate of the number of tourists						
Number of tourists	17000	21530	20000	31630	10000	19680

Comparison result			
	Result1	Result2: $\overline{\alpha_{43}} = (0,0,0)$	Result3: $\overline{\alpha_{41}} = (0,0,0)$
X1	21530	21530	19420
X2	31630	31630	32630
X3	19680	43970	43970
Economic	8374290	9423879	9423879

The results obtained are shown in the table, where the coefficients are based on publicly available data from the Venetian government. At the same time, in order to represent the key variables, such as the maximum number of visitors to Piazza San Marco, we changed some of the variables, and the results were different by limiting the number of visitors to Piazza San Marco for day trips and the number of visitors to Piazza San Marco for day trips.

The fitting results show that the optimal number of tourists coming to Venice every day is 73,020, including 21,530 tourists staying in hotels, 31,630 people staying in non-hotels such as B&B, and 19,680 tourists visiting day trips, and the tourism industry in Venice can earn 8374290 euros per day.

Due to space constraints, data for outcomes 2 and 3 are not presented here. From this result, we can conclude that economic and environmental measures are the most important in different locations due to their geographical location, the different ways people choose to travel and stay, and the different degrees of environmental impact.

4.2 How to Balance Attractions with Fewer Tourists

4.2.1 Calculation between Hypothetical Location

We aim to measure and balance the sustainable development capacity of three locations: A, B, and C. B has the highest number of tourists, leading to greater infrastructure pressure and carbon emissions. A and C are less popular with fewer tourists. Model 1 helps compare the optimization results for the total population per year, with the left figure showing B's fitting optimization and the right figure comparing all three locations.

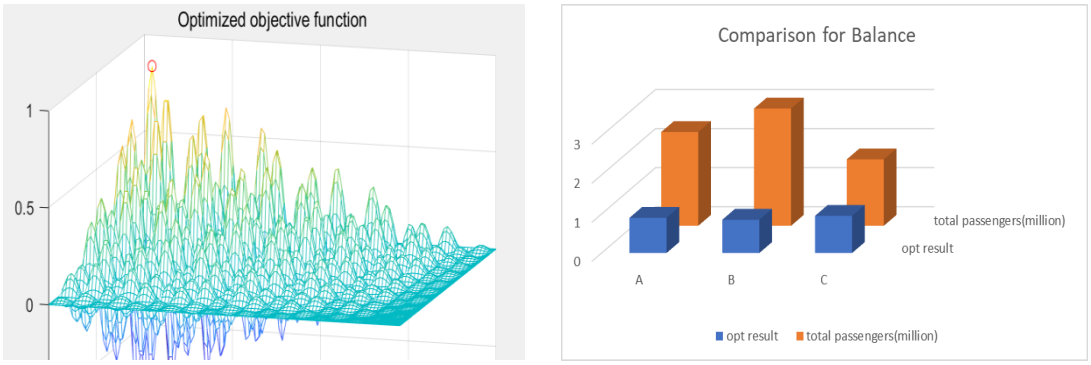


Figure 15: left: B's optimization, right: comparison of all three

In addition, in order to balance attractions A and B, we merge attractions A and B, and optimize them using model 1 by multiplying the correlation coefficients of the different volumes of A and B, and the method and the results are as follows:

$$M = \frac{M_1}{M_1 + M_2} M_1 + \frac{M_2}{M_1 + M_2} M_2$$

4.2.2 Result and Conclusion

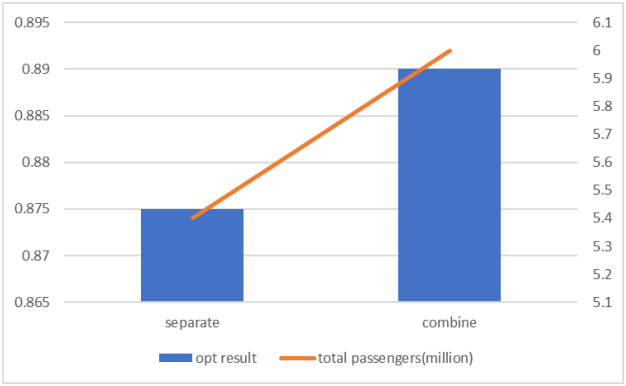


Figure 16: A and B binding analysis compares with A and B separate analysis

Although B attracts more tourists, A and C show greater sustainable development benefits. Limiting taxes and visitor numbers may slightly reduce economic output but lower emissions and infrastructure pressure while increasing resident satisfaction. Sharing tourists between A and B yields better results than managing them separately, offering a solution for balanced development..

5 Sensitivity Analysis

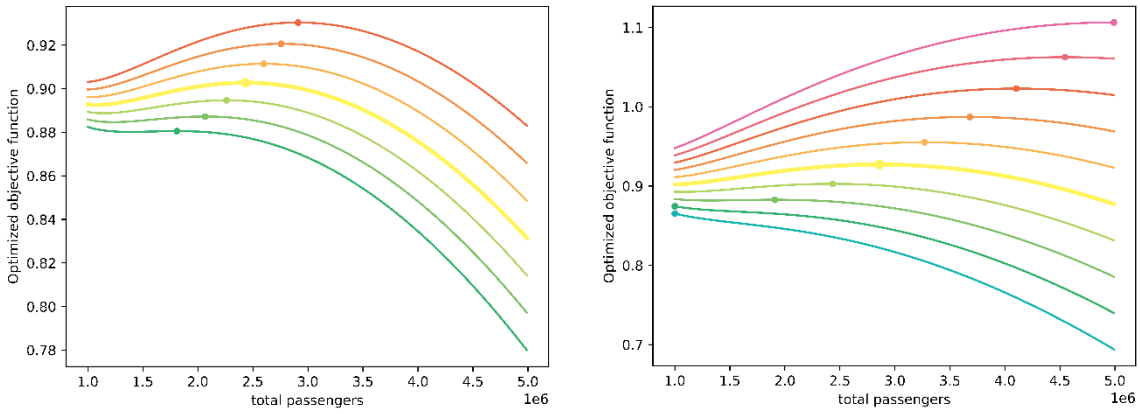


Figure 17: The impact of model coefficient adjustment on coefficients

We analyzed the sensitivity of the key parameters to the resulting x_{opt} in our EES model, paying special attention to the factor demonstrating how tax rate influences the number of tourists, recoded as TIT in figure 17, the value of k and b mentioned in 3.2.2-1, the weight coefficients mentioned in 3.4, carbon footprint emission coefficient and social impact factor λ.

To tell the sensitivity of these parameters in our EES model, we use the formula:

$$y = \frac{param}{x} \cdot \frac{\Delta x_{opt}}{\Delta param}\bigg|_{x_{opt}=1}$$

The second term in this formular demonstrate how x_{opt} would change if you change the value of the parameter, and the first term is used to avoid the impact of the magnitude of the parameter itself.

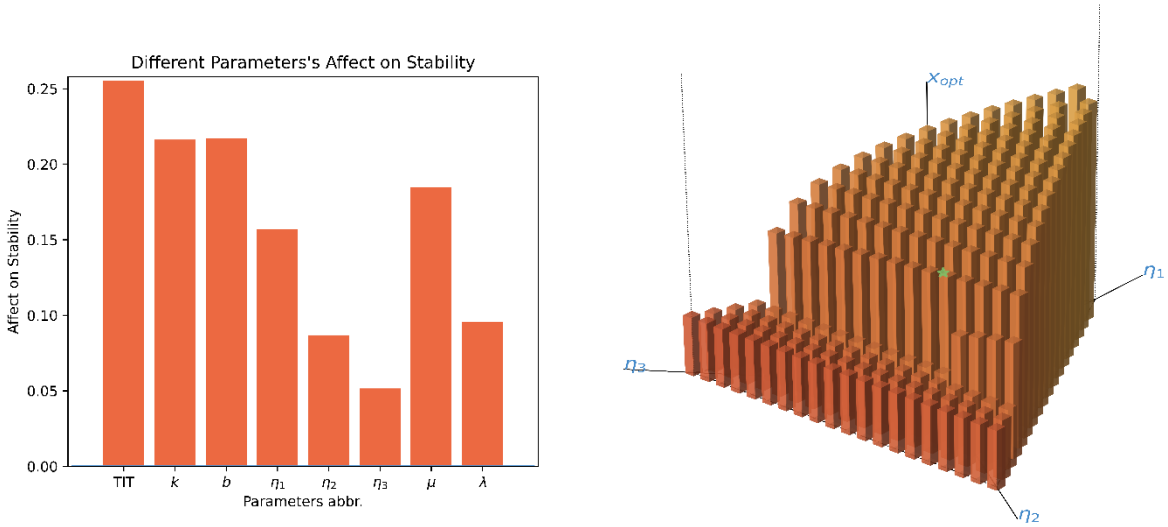


Figure 18-19: The Influence of Different Coefficients on Model Stability

Through simulation and calculation, we got how sensitive the model is when talking about different parameters. This is shown in figure 17. These results aren't greater than about 1/4, which demonstrates that our model is quite stable. TIT has the highest impact on the stability of the model and η_3 has the lowest impact on the stability of the model. When migrating the model to other circumstances, we should focus more on the way tax rate influence the number of tourists in that place, and focus less on some other parameters in the model.

Figure 17 shows how changing μ and λ would affect the rating function. Figure 18 shows how changing η_1, η_2, η_3 would affect x_{opt}.

6 Conclusion

6.1 Model Strengths

1、Generalizability: The model exhibits strong predictive accuracy across diverse regions, demonstrating its robust design for sustainable planning, with the exception of Juneau.

2、Prehensiveness: By integrating economic, social, and environmental factors, the model optimizes these aspects, establishing a robust foundation for sustainable site development.

3、hodological Rigor: The model employs a sophisticated SSA methodology, enhanced by FCLP updates, ensuring the integrity and reliability of its calculations.

4、Sensitivity and Adaptability: The model demonstrates high sensitivity to external factors, such as economic policy shifts and emergencies, enabling it to make timely and effective predictions while adapting to dynamic conditions to support government-led sustainable development initiatives**.**

6.2 Model Weakness

The model's reliance on historical data can introduce uncertainty and potentially compromise prediction accuracy. So increased data availability will facilitate more comprehensive sustainability planning, including precise crowd management, thereby advancing sustainability efforts.

6.3 Brief Summary

Our model scientifically and organically integrates the three quantitative evaluation criteria of sustainable tourism. By collecting relevant data and adjusting parameters in different locations, we can obtain data on taxes and the most likely number of tourists suitable for tourist areas. It is worth mentioning that the figure below is a collection of factors of relevant tourism stakeholders that we have sorted out, so we also hope that local governments and relevant departments can refer to it.



Figure 20: Overview of Tourism Development Stakeholders

7 Memo to Tourist Council of Juneau

Date: January 27, 2025

To: Leadership, Travel Juneau

From: ICM Team # 2508991

Subject: A Three-dimensional Sustainable Tourism Assessment Model——EES

To whom it may concern,

To address your inquiry about sustainable tourism in Juneau, we assessed its economic, ecological, and social dimensions, considering Alaska's unique tourism resources. Utilizing data from official sources, we analyzed recent trends and developed optimization strategies. Our suggestions and projections are outlined below.

- **Projections:**

1. **Continued Growth:** We anticipate a sustained increase in annual tourist arrivals in Juneau, driven by robust economic growth. This trend is likely to persist in the near term.
2. **Socio-Environmental Challenges:** Despite the economic benefits, the growing tourist population and climate change will continue to pose significant challenges to the local environment and community. However, based on our current data, the impact on residents' quality of life remains manageable.

- **Recommendations:**

1. **Managed Growth:** A slight increase in tourist numbers is acceptable in the short term. Natural constraints such as distance, seasonality, and transportation limitations will help regulate visitor influx, and tourism can continue to be a major economic driver.
2. **Proactive Measures:** While advocating for a more permissive approach, we strongly recommend implementing robust regulations, establishing a maintenance office, and enhancing social safety nets to minimize the environmental footprint and social disruptions caused by tourism.
3. **Contingency Planning:** If the annual tourist arrivals exceed the optimized threshold of 2.4 million, stricter regulations should be enforced. Additionally, relevant departments should closely monitor the situation and develop contingency plans.

- **Model Characteristics:**

1. **Key Variable Focus:** Our model effectively captures and optimizes the relationship between tourist numbers and tax rates.
2. **Quantifying Intangibles:** We have successfully quantified relatively subjective concepts like "perceived impact" and "ecological impact" using a robust methodology, allowing for a more scientific and accurate assessment.
3. **Data-Driven Approach:** Our model is grounded in a comprehensive dataset sourced from reliable government platforms, international organizations, and peer-reviewed literature.

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